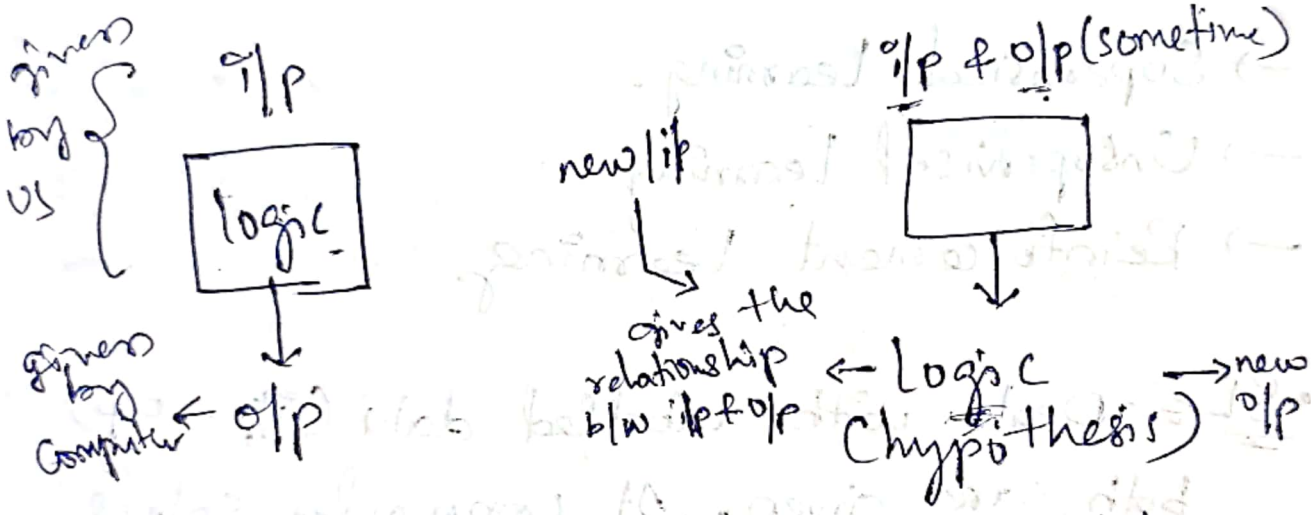
Class 1

Machine Learning

ML is a part of Computer Science where we teach computer's to learn from data, find patterns & make decisions/predictions.

Traditional Computing Vs ML algorithm



→ we don't teach the ML by giving idea or logic code, we just provide data and machine finds the logic / relationship b/w i/p & o/p

eg: spam emails

TC	ML
→ check every email & find patterns then apply logic	→ feed data & automatically detects patterns

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→ after finding the patterns it decides that the email should be in spam or not spam.

eg:- Weather prediction, Medical diagnosis youtube recommendations, Movie recommendation on OTT's etc... & many more.
face Recognition & Self driving Cars

ML Types

→ Supervised Learning
→ Unsupervised Learning
→ Reinforcement Learning

• **SL** :- Deals with labelled data (i/p → o/p) both are given. It commonly solves two types of problems

(grade spam/notspam cats/dogs) • Classification (Categories) finite set of defined o/p classes

• Regression (Continuous values) marks eg: aqi / house price / stock price

Labelled data : It is a raw data (such as images, text, or audio) that has been paired with a meaningful tag, label, or annotation that informs a ML model about the desired outcome

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• UL : Deals with unlabelled data, we give only i/p & ML model identifies patterns using only i/p data. It commonly solves clustering & association problems.

• clustering

eg: In amazon ppl when they buy iphone they also buy a charger. So when a new user buy's an iphone we can suggest a charger. | BMI of alcoholic & non alcoholic persons

→ We make cluster's of related data

• RL : goal oriented learning, we use RL when there is a well defined target to acheive.

eg: • Designing a game, Robotics, self driving Cars.

* Tracing a pet

→ Commonly used with SL | UL.

→ Works with rewards & penalties.

*

→ We use commonly SL mostly.

→ Coz machines loves labelled data

→ They find the patterns & make better predictions by SL.

→ SL is widely used in the Software Industry.

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Supervised ML

SL algo's trains mdl's on labeled datasets, meaning each input has a known correct output, to learn patterns & predict outcomes for new data.

Eg: Dataset

Email Text	Sender email	Link	no. of Exclamations	Label
"Win a brand new iphone"	promo@site.com	yes	3	spam
"Dear, Temple final details"	manager@company.com	no	0	notspam

Input values (X) — o/p (y)

→ for single input & o/p x, y, multiple X, Y

→ we give / feed the 2D data to model.

→ every row represents single data point

→ all the columns in our dataset is known as features

→ features in ML gives valuable info to build relationships

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Features

* Independent → input values (X)
* Dependent → output (y)

→ output is always dependent on the input features.

→ Working of ML pipeline

① Data → preprocess/clean → feature engi

Feature engineering

- Remove unnecessary features
- Add new features based on old
- Encode data to numeric

→ machine Learning loves numerics values.

② select Model

→ every algorithm uses different type of logic & maths behind the scenes. So selecting correct model for the data is necessary.

→ Based on problem we select model

③ Train/Test /Split.

- we split data into two ~~types~~ categories/parts

① Training - 80% (mostly)

② Testing - 20% data (Testing)

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→ We split the data, shuffle the data

→ We train the ~~data~~ model on training set & test the ~~da~~ model on testing part of data

→ Before real world predictions, to check how well is model trained on the data.

→ In test data we have output also & the output is correct, so we can easily test and evaluate the performance of our model.

Note: We don't know the correct o/p on real world data, so that's why we divide the data into training & testing.

(4) check whether it is underfit / overfit (model evaluation)

SL - Linear (Regression)

- output is a continuous numerical values
- function => Relationship b/w the i/p & o/p
- In linear Regression the function is a straight line & the equation is

$$\boxed{y = mx + c}$$

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→ Here we know y (output) & x (input) we have to find 'm' and 'c', to get the relationship between input & output.

Classification

→ output is a category or class

eg: • spam/not spam
• cat/dog/rabbit
• Cancer is on which stage.

→ If there are only 2 possible outputs or classes, then it is known as binary classification.

→ If we have more than 2 possible outputs classes/categories then it is known as multi-class classification.

→ Multi-label calassification:
If we give multiple labels to a particular value then it is multi-label classification.

eg: for the image of cat we can give a cat label as well as an animal label.

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